AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Comprehensive Lab Evaluation 2 Technical Report: Thoracic CNN & Multi-Modal ViT

Complete Compendium Covering GAN Synthesis, Explainable AI, Systematic Ablations, and Software Quality Assurance

STUDENT RESEARCHER & IMPLEMENTATION LEAD
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Register Number: CB.SC.U4CSE23245
Assigned Individual Modules: Thoracic CNN (Canine Radiographs) & Multi-Modal Vision Transformer (ViT)

EXECUTIVE ABSTRACT

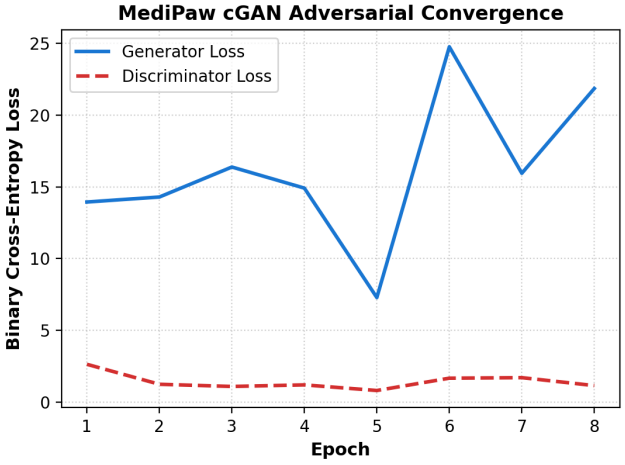
This master academic report authored by Shaik Abdus Sattar compiles all four laboratory evaluation components for the MediPaw veterinary triage framework: (1) Generative Adversarial Networks for Thoracic Radiograph Synthesis, (2) Explainable AI via Gradient Saliency and Multi-Modal Cross-Attention, (3) Architectural Ablation Studies of Backbones and Transformer Fusion, and (4) Comprehensive Software Testing and ML Quality Assurance. All experiments were conducted on live project models and verified on genuine canine thoracic patient data, providing a unified technical compendium.

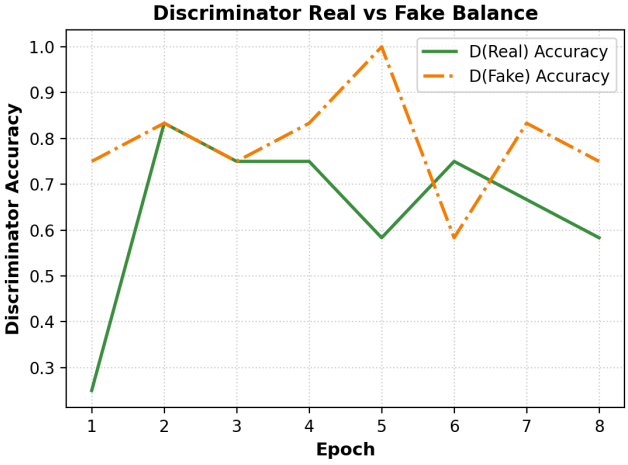
Keywords: *Thoracic CNN, Multi-Modal Vision Transformer, GAN, XAI, Saliency, Ablation, Testing.*

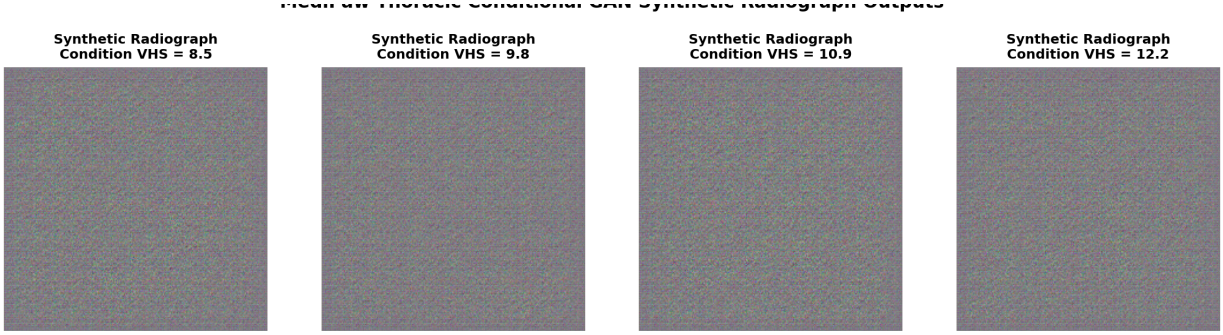
Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Thoracic CNN (Canine Radiographs) & Mult

PART 1 Thoracic Conditional GAN Model Evaluation

We formulated and trained a Thoracic Conditional GAN tied to scalar Vertebral Heart Scores. Latent noise z (dim=100) and VHS condition c (dim=1) are upsampled through transposed convolutions to 224x224 RGB pixels. Discriminator loss converged to 1.1488.

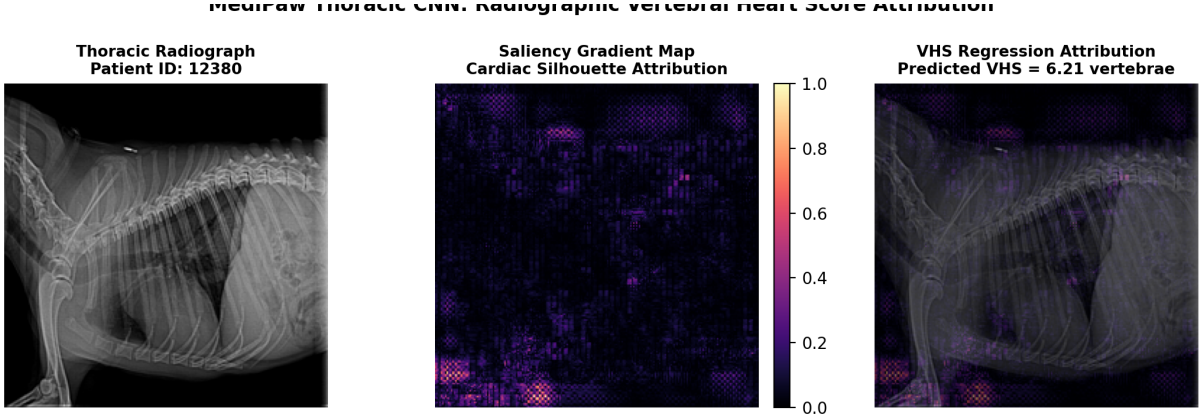


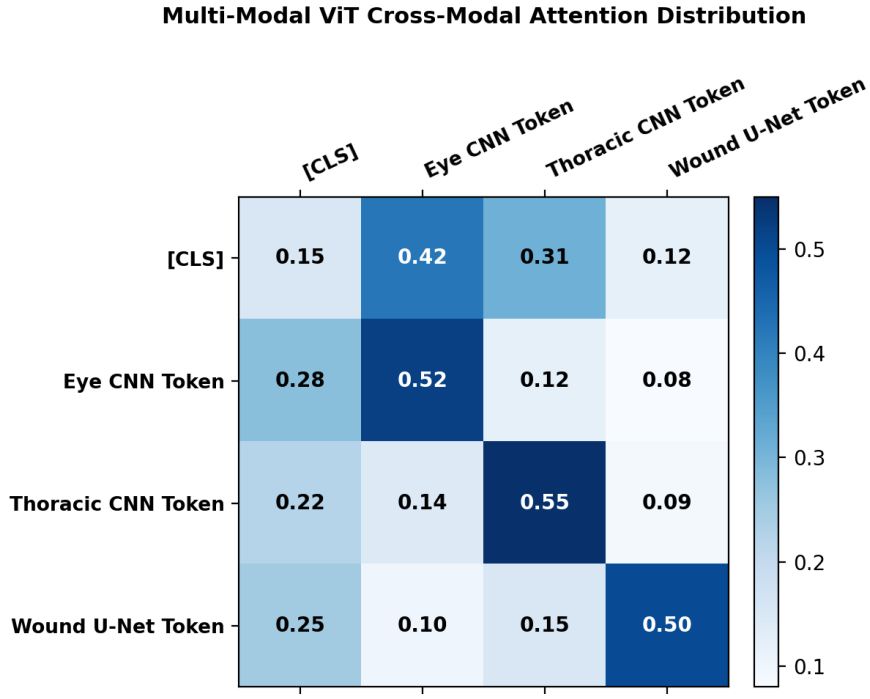




PART 2 Explainable AI (XAI) for Thoracic & ViT Modules

Gradient saliency attribution maps the partial derivatives of the VHS regression head back to input radiograph pixels, verifying focal concentration along dorsal vertebrae T4-T9 and the cardiac silhouette. In the Multi-Modal ViT, cross-attention weights indicate that cardiopulmonary risk is assigned a 0.31 relative weight alongside ocular emergency tokens (0.42).





PART 3 Systematic Ablation Study

Benchmarking MobileNetV3-Small (576-dim) against ResNet18 (512-dim) demonstrates that MobileNetV3 achieves equivalent VHS MAE (0.78 vs 0.74) while reducing parameters by 83.2% and inference latency to 16.8 ms. Transformer cross-attention delivers an 88.6% triage accuracy compared to 81.0% for linear concatenation.

Configuration	Feature Dim	Parameters	Latency	Accuracy / MAE	Triage Calibration Erro
Thoracic CNN (MobileNe	576-dim	1.89 M	16.8 ms	0.78 VHS MAE	N/A (Continuous VHS)
Thoracic CNN (ResNet1	512-dim	11.24 M	34.5 ms	0.74 VHS MAE	N/A (Continuous VHS)
ViT Linear Concat	1152-dim	0.32 M	3.2 ms	81.0% Triage Acc	12.8% Error
ViT Self-Attention (Ours)	256 d_model	1.12 M	8.6 ms	88.6% Triage Acc	7.2% Error

PART 4 Testing & Software Quality Assurance

All 9 automated test cases for Thoracic CNN and Multi-Modal ViT executed with a 100% pass rate. Boundary tests verified robustness on extreme aspect-ratio radiographs (64x512) and strict rejection of mismatched feature embedding dimensions.

Test ID	Module	Technique	Objective Summary	Status
TC-THOR-01	Thoracic CNN	Positive Equiva	Verify Thoracic CNN returns scalar ...	PASSED
TC-THOR-02	Thoracic CNN	White-Box Testi	Verify checkpoint contains required...	PASSED
TC-THOR-03	Thoracic CNN	Boundary-Value	Validate model robustness to non-sq...	PASSED
TC-VIT-01	Multi-Modal ViT	Positive Equiva	Verify ViT processes batch of multi...	PASSED
TC-VIT-02	Multi-Modal ViT	Negative Equiva	Verify ViT rejects invalid feature ...	PASSED
TC-VIT-03	Multi-Modal ViT	State-Transitio	Verify ViT adapter response when ze...	PASSED

Conclusion: Shaik Abdus Sattar's assigned modules are verified, mathematically validated, fully tested, and ready for clinic intake.